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Leetcode 512 : Game Play Analysis II.

Our goal is to understand how to retrieve the device used by each player on their 1st login date.

We are given one table : Activity.
It contains 4 columns.

column	meaning
player-id	unique id of the player
device-id	device used by the 〃.
event-date	date the player logged in.
games-played	number of games played.

what do we need to find?

↳ for every player, find the device they used when they logged in for the very 1st time. ___/___/___

→ we need the device from the earliest login, not the latest login.

example →

player-id	device-id	event-date	games-played
1	2	2016-03-01	5
1	2	2016-05-02	6
2	3	2017-06-25	1
3	1	2016-03-02	0
3	4	2018-07-03	5

player 1 • 1st login: 2016-03-01 → device 2
2nd " : 2016-05-02 → device 2
answer: device 2.

player 2 : only one login: 2017-06-25 → device 3.
answer: device 3.

player 3 : 1st login: 2016-03-02 → device 1
2nd login: 2018-07-03 → device 4,] answer: device 1.

output :-

player-id	device-id
1	2
2	3
3	1.

* find the earliest login date for every player,
  ↳ MIN()

* find the device used on that earliest login date

```
SELECT
    a.player_id,
    a.device_id,
FROM Activity a
JOIN (
    SELECT
      player_id,
    MIN(event_date) AS first_login
    FROM Activity
    GROUP BY player_id
) b
```

→ final select.

→ we display only the two columns requested in the question.

→ we have 2 table.
↳ original activity table
↳ result of our inner query.

→ we are joining them to get the device ID associated with the 1st login.

→ finds the earliest date from the player login with column name first_login.

↳ imagine we have multiple records belonging to player 1, sql groups all records belonging to player 1 together.

~~the~~ the inner query gives us only the player ID & 1st login date.
It doesn't give the device ID.
That's why we need join.

ON a.player_id = b.player_id → match records belonging to the same player.
AND a.event_date = b.first_login;

↓
bcz one player can have multiple login records
we only join the record whose event date equals that player's earliest login date.

for ex, player a takes a should match player 1 in b-

for player 1:-
2016-03-01 = 1st login → match ✓
2016-05-02 = not 1st login → no match

now we have the correct row containing the device ID.

Tracing player 2,

inner query result,
MIN (event_date) = 2016-03-02.

Join comparison

original date	first login date	match?
2016-03-02	2016-03-02	✓
2018-07-03	2016-03-02	✗

Result ✓